

Indian Institute of Technology Gandhinagar

CE 302: Structural Analysis

Spring 2026

Instructor	Manish Kumar (mkumar@), S. Chandrasekaran Office: AB 5/333
TA	TBD
Timings	Lecture: M (10:00–11:20 AM), Th (8:30–9:50 AM); Tutorial: T (8:30–9:50 AM)
Venue	Lecture: AB 7/101 (M), AB 7/210 (Th); Tutorial: AB 7/101 (T)
Credits	3-1-0-4
Prerequisite	ES 221 (desirable)
Course content	Degrees of indeterminacy (flexibility & stiffness); Determinate and indeterminate structures: trusses, cables, arches, beams and frames; Deflections: moment area theorem, conjugate beam method, principle of virtual work, energy method, Castigliano's theorems, unit-load and unit-displacement theorems, Maxwell's theorem, Betti's law, method of consistent deformations; Displacement method of analysis: slope-deflection method, moment distribution method, column analogy method; Influence lines, Muller-Breslau's principle; Introduction to matrix method; Introduction to a structural analysis software program
Text/references	1. <i>Structural analysis – 10th edition</i> by R. Hibbeler (<i>Textbook</i>) 2. <i>Fundamentals of structural analysis</i> by K. Leet, C. Uang, and A. Gilbert 3. <i>Fundamentals of structural analysis</i> by H. West, and L. Geschwindner 4. <i>Structural analysis</i> by A. Kassimali
Grading policy	Mid-term exam: 20 End-term exam: 30 Project: 15 Quizzes: 10 Assignments: 10 Class participation: 10 Activity: 5 ◦ <i>The letter grade 'A' is guaranteed at scores of 80 and above.</i> ◦ <i>The letter grade 'F' shall be awarded at scores below 40.</i> ◦ <i>Participation in interactions and activities during lectures enhance understanding, interest and long-term retention of the concepts. Accordingly, a class participation score of 10 shall be assigned for active participation in all lecture sessions. A relaxation for up to three lecture sessions may be considered on account of medical or personal situations. The score shall be zero in the absence of the sustained class participation.</i> ◦ <i>Final letter grades will be based only on the two exams (40% each) and the second quiz (20%), if a student is absent in more than two tutorials.</i> ◦ <i>A relaxation in the participation requirements noted above is possible in serious medical situations (e.g., long hospitalization) certified by the Medical Center of IIT Gandhinagar.</i>
Activities	Some additional activities (e.g., site visit, guest lecture, special quizzes) may be planned, credits for which may be adjusted from the final exam or participation score.
Assignments	Assignments will be due at the beginning of the respective lecture/tutorial. These will be graded only when (1) submitted by due time, and (2) prepared on A4 sheets with hand-written work in pencil.
Project	The project will be performed in a group. It will involve planning, fabrication and experimental testing of a specimen, and comparison of observations with analytical results.
Punctuality	Entering late in the class shall not be permitted.
Honor code	Honor code shall be enforced in accordance with the Institute rules.

Learning outcomes: At the end of this course, a student should be able to (1) *at least* calculate deflections and internal stress resultants, namely, axial force, shear force and bending moment, for determinate trusses, beams and frames, (2) analyze statically indeterminate structures, and (3) analyze a structural system using a commercially available software program. The student should also be able to develop some appreciation towards modelling actual structures.

Course plan: A tentative schedule of topics to be covered is noted in the table below. There will be an assignment or an activity, on an average, every week.

Week	Topics
1	Introduction to structural analysis: structures, loads, load path, idealization of structural system, tributary areas, support conditions; Static determinacy, stability
2	Application of equilibrium equation on simple systems; Analysis of statically determinate trusses: method of joints, method of sections
3	Internal stress resultants (shear and moment diagrams) for statically determinate beams and frames, method of superposition
4	Cables and arches, influence lines, Muller-Breslau's principle
5	Approximate analysis of indeterminate frames: portal and cantilever methods
6-7	Elastic beam theory, Deflection: double integration method, moment-area theorem, conjugate beam method
7-8	Deflections: energy methods, principle of virtual work, Castiglano's theorem, unit load and unit displacement methods
9	Force method of analysis, Maxwell's theorem of reciprocal displacements, Betti's law, Qualitative influence lines
10-11	Displacement method of analysis: slope deflection equation, moment distribution method, column analogy method
12-13	Kinematic determinacy, analysis of truss, beam and frame using stiffness method